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Received: 27 February 2023 | Revised: 10 July 2023 | Accepted: 3 August 2023

DOI: 10.1002/gdj3.217

Geoscience Data Journal | RMetS | WILEY

DATA ARTICLE

# A 5-km gridded product development of daily temperature and precipitation for Bangladesh, Nepal, and Pakistan from 1981 to 2016

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**Funding information**
Asia-Pacific Network for Global Change Research

## Abstract

Many efforts have been made by the scientific community to produce gridded datasets with high spatial resolution because they are essential for climate change assessment, impact studies, decision-making, etc. This study fits into this context and describes the methods used to prepare a 5-km gridded product of precipitation and minimum and maximum temperatures by merging observed data from meteorological stations, from 1981 to 2016, of Bangladesh, Nepal, and Pakistan

**Dataset** Identifier: https://doi.org/10.6084/m9.figshare.21565149.v1

Creator: Ali Shaukat, Reboita Michelle

Title: 5-km gridded datasets of precipitation and minimum and maximum temperatures for Bangladesh, Nepal and Pakistan

Publisher: figshare.com

Publication year: 2022

Resource type: Netcdf Data Files and Metadata

Version: 1.0

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with ERA5 reanalysis. The step-by-step methods for station data quality control and the development of the 5-km gridded data are presented. Additionally, we use the 5-km dataset to show the main climate features of the three countries, which facilitate comparison with other data sources in the literature.

**KEYWORDS**
climate change, climate variability, gridded data, high-resolution

## 1 | INTRODUCTION

Climate research relies heavily on primary variables such as observed precipitation and minimum and maximum temperatures, which are recorded at meteorological stations located nonuniformly and dispersedly across countries (Hussain et al., 2022; Khan et al., 2022). However, these features make it difficult to conduct studies that require high-resolution observed gridded datasets, such as assessment studies, numerical models' validation, statistical downscaling, and policy development for health, agriculture, and water resources.

With the onset of the Anthropocene, the numerous fields of study that investigate the effects of climate change require spatially and temporally consistent climate data at high spatial and temporal resolutions (Bjorkman et al., 2018; Hewitt et al., 2017; Trisos et al., 2020). In response to this need, an ever-growing number of climate datasets have been created (Abatzoglou et al., 2018; Beyer et al., 2020; Fick & Hijmans, 2017; Karger et al., 2017; Navarro-Racines et al., 2020) making use of observations, reanalysis products, climate model outputs, or some combination thereof. Climate reanalysis products represent a major achievement of climate science (Buizza et al., 2018). These products optimally combine a wide range of surface and satellite observations with a dynamical model in order to produce a self-consistent dataset.

Despite the advantages of reanalyses, there is limitation of coarse spatial resolution (Hersbach et al., 2020; Sabater, 2017). Only APHRODITE provides climatology (30 years) at a resolution of 0.05° for Asia, hence lacking the daily data. ERA-Land is developed at a finer resolution (0.1° × 0.1°) but it is a dynamical downscaled product of ERA5. Against this backdrop, this study presents the methodology used to prepare a 5-km gridded product of maximum and minimum daily temperature and precipitation dataset for Asian-Pacific countries: Bangladesh, Nepal and Pakistan, under the Asian-Pacific Network for Global Research (APN) project named "Towards robust projections of climate extremes and adaptation plans over South Asia" (https://doi.org/10.30852/p.4583).

## 2 | METHODOLOGY

The methodology presented here to get the high-resolution dataset is based on a combination of different methods used in the literature to developed gridded datasets. We highlight that one crucial step is the quality control of the observed datasets.

### 2.1 | Study area

The study area comprises Bangladesh, Nepal, and Pakistan (Figure 1), which are collaborative countries in the APN project. Pakistan has a very steep topography in the north of the country, where the Karakoram and Himalayas mountains (reaching 8,000 m) are located. Bangladesh's terrain is generally flat, except for a few mountains in the north. Nepal is a landlocked South Asian country with a mainly hilly topography, with elevations ranging from 60 m in the south to the world's highest peaks, including Mount Everest (8,848 m).

### 2.2 | Data

Minimum and maximum temperatures and precipitation data from meteorological stations were acquired from the meteorological departments of Bangladesh, Nepal, and Pakistan.

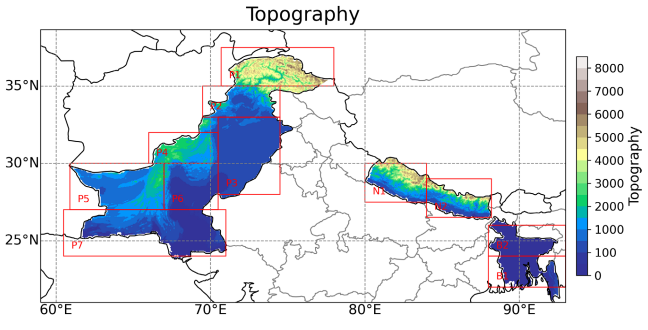



FIGURE 1 Topography with 5 km of grid resolution obtained from GTOPO30 (2022) for Bangladesh, Nepal, and Pakistan, and the red boxes show our analyses regions.

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ERA5 reanalysis, obtained from the European Center for Medium-Range Weather Forecasts (ECMWF – https://cds.climate.copernicus.eu/, Hersbach et al., 2020), was used to merge with the observations for the period of 1981–2016. The variables, air temperature at 2 meters and precipitation, were obtained with a 1-h frequency and a horizontal resolution of 0.25°. The values of minimum and maximum temperatures were extracted from the diurnal cycle of the ERA5 reanalysis. The hourly precipitation was accumulated in 24 h (from 0300 Z to 0300 Z) to standardize with the observed data.

The temperature from the stations and reanalysis was spatially interpolated for the gridded data generation. The temperature was adjusted after the interpolation process to consider the influence of topography (lapse rate effect; Section 2.3.2). For this purpose, we used a digital elevation model (high resolution) – Global 30 Arc-Second Elevation (GTOPO30) from the U.S. Geological Survey (USGS – https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-global-30-arc-second-elevation-gtopo30). GTOPO30 is a global raster digital elevation model (DEM) that provides terrain elevation data with a horizontal grid spacing of 30 arc seconds (approximately 1 km). GTOPO30 was derived from several raster and vector sources of topographic information.

## 2.3 | High-resolution grid data construction

### 2.3.1 | Station data quality control

The first step of this study involved applying quality control methods, both objective and subjective, to the time series. For the objective method, statistical analyses were performed to identify outliers and validate the consistency of spatiotemporal data. Initially, outliers were identified using simple statistical analysis, such as a box whisker plot. However, immediate exclusion of outliers was not done as they may represent extreme events. The objective method further examined spatiotemporal data consistency. Temporal consistency is based on the concept that an individual observation should be statistically comparable to observations made at the same time (e.g. day or month) in other years. On the other hand, spatial data consistency uses nearby observations with adequate data to estimate the value at a target station (Xu et al., 2014).

In this study, we applied the spatiotemporal consistency method developed by Xu et al. (2014), which is a probabilistic spatiotemporal approach based on a spatial regression test to assess uncertainty quantitatively in observations at meteorological stations. However, in some cases, even after applying the aforementioned procedures, the algorithm is unable to determine if a suspicious result is real or spurious. Consequently, the researcher performs the subjective task of deciding whether to retain or discard the data (Rozante et al., 2018).

### 2.3.2 | Steps of the 5-km grid dataset preparation

To prepare a grid dataset with a 5-km horizontal resolution for precipitation and minimum and maximum temperatures, we merged observed data (after the quality control) with ERA5 reanalysis, which is a regularly spaced dataset. The steps for preparing the spatialized grid at 5-km resolution are described below and summarized in Figure 2 for Pakistan.

a. Data Organization: To illustrate the preparation of the 5-km data, we begin by showing Figure 2a, which displays an ERA5 variable (precipitation or temperature depending on the analysis) on the background and the location of the stations (black dots). The first task is to remove ERA5 grid points that are closer to the stations. Then, a surrounding distance of 100 km is calculated from each station's location and displayed in grey boxes in Figure 2b. According to the World Meteorological Organization (WMO), weather conditions recorded at a meteorological station can be considered representative up to a radius of 100 km. The ERA5 grid points inside the grey boxes are eliminated from the dataset to give high influence to the observed data during spatial interpolation. Following that, the dataset is tabulated in the manner shown in Figure 2c.
b. Spatial Interpolation: The tabulated precipitation data (Figure 2c) are interpolated using the ordinary Kriging method (Figure 2e) at a 5-km regular grid (Figure 2h). The steps for preparing the high-resolution precipitation dataset are illustrated in Figure 2 with blue arrows. Temperature requires an additional step, which is explained further below.

### 2.3.3 | Additional procedure for air temperature

In the case of temperature, an additional step is included in the technique to reduce errors during interpolation in regions with a scarcity of observed data and rugged topography. As a result, the temperature is adjusted by considering the estimated average of the Lapse Rate of Temperature

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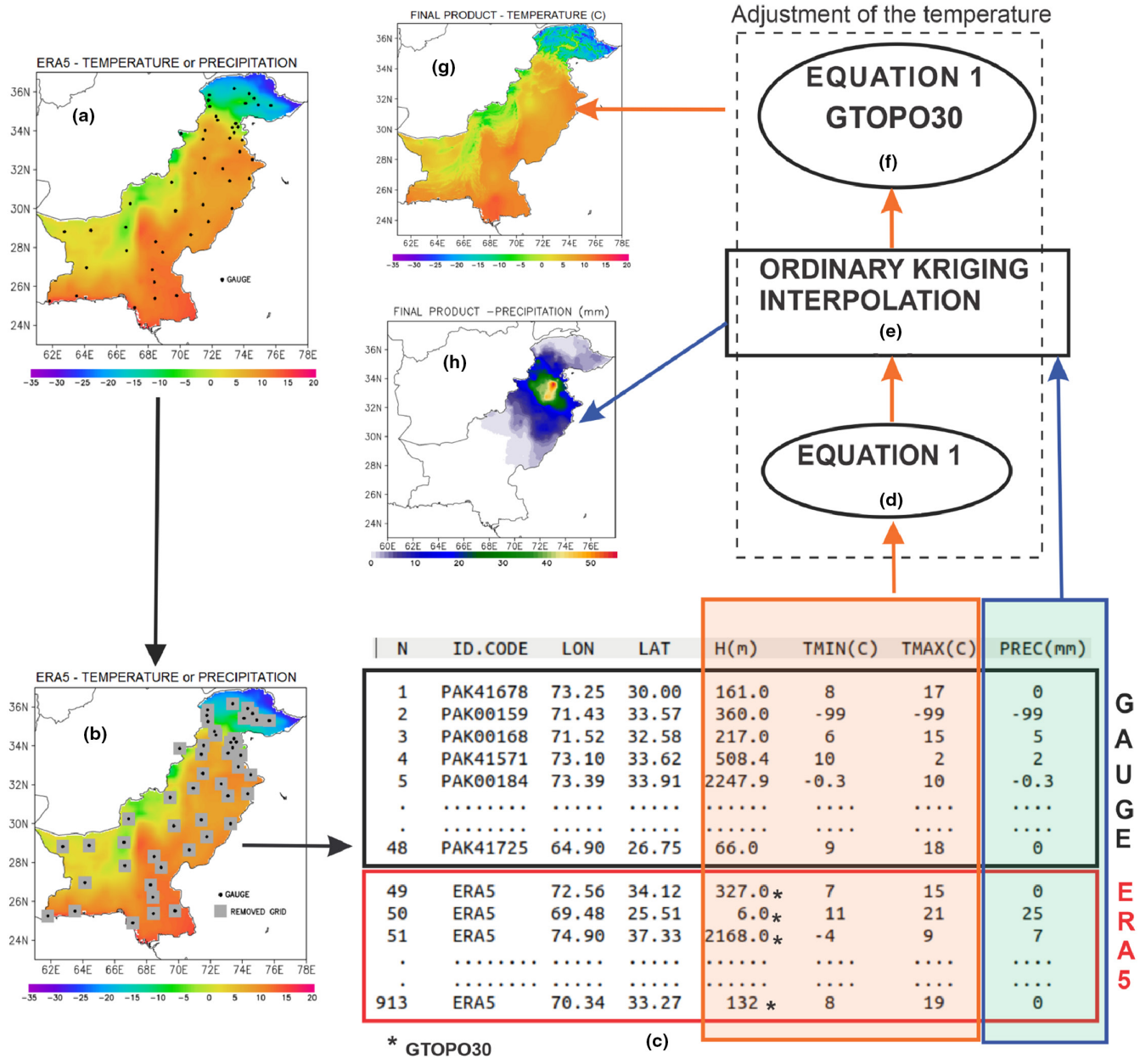



**FIGURE 2** Schematic illustration of the steps to obtain the 5-km grid dataset product.

(LRT). To this end, the average values of LRT for Pakistan and Nepal were calculated. However, this step was not necessary for Bangladesh due to its flat topography.

The LRT is estimated using a simple linear regression equation (Equation 1; Blandford et al., 2008; Minder et al., 2010) with station mean temperature climatology.

$$T = Co + LRT^{*}H \quad (1)$$

where T is a dependent variable and represents the mean temperature (°C), H (km) is an independent variable that represents the height in kilometres, Co is a linear intercept (°C), and LRT is the temperature lapse rate (°C/km).

After obtaining the LRT, the 5-km grid temperature dataset is generated following Section 2.3.2 and the steps outlined below.

a. The temperature data (from the table in Figure 2c) are standardized to sea level using the LRT and elevation (H) (Figure 2d), and, in the sequence, interpolated to a 5-km resolution using the ordinary Kriging method (Figure 2e).
b. To include the topography effect on the temperature on the 5-km gridded dataset, Equation 1 and the topography from GTOPO30 (Figure 2g) are used.

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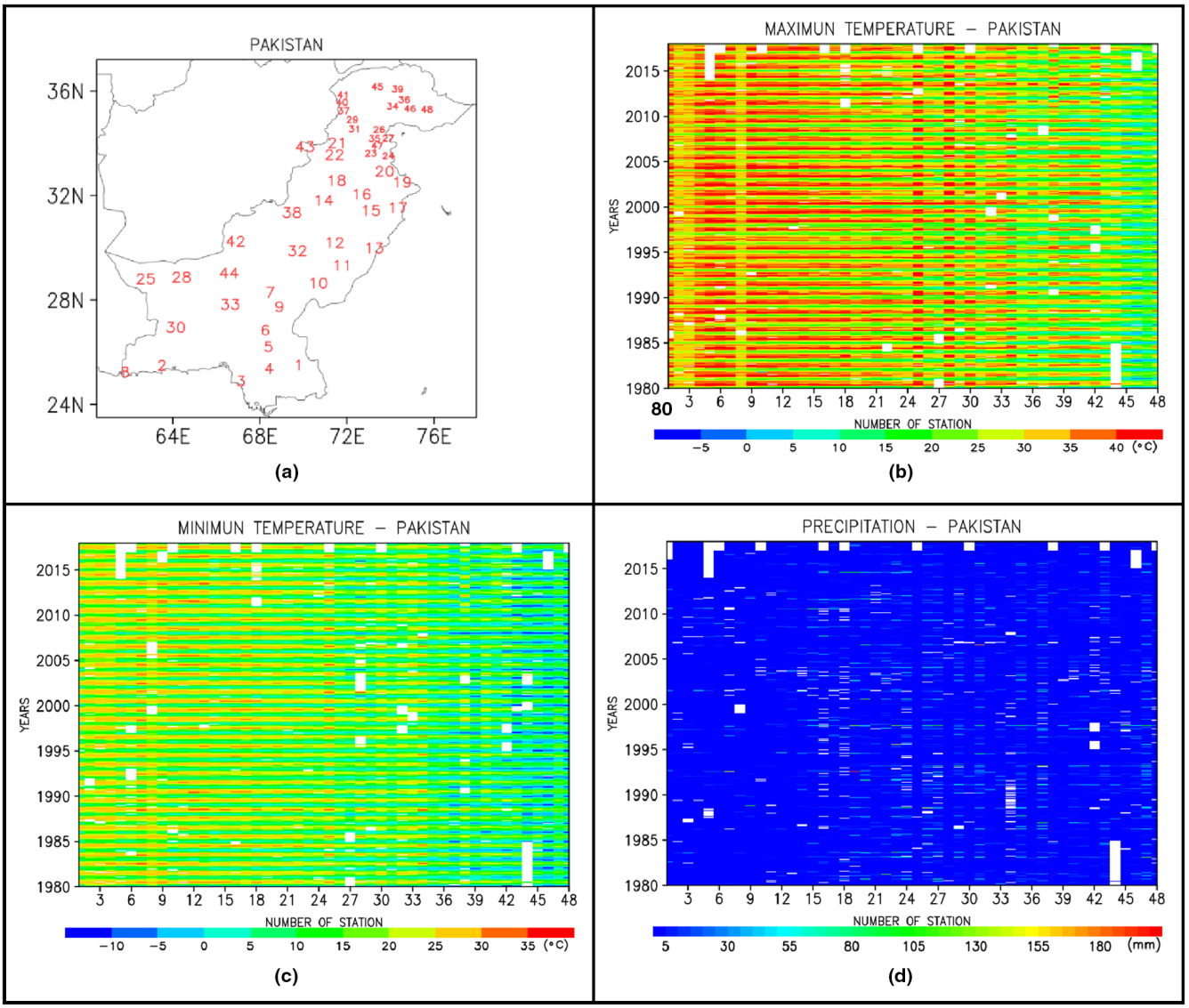



FIGURE 3 Spatial distribution of stations (a), TMAX valid points (b), TMIN valid points (c), PREC valid points (d) for Pakistan. White squares are missing data. The number of stations was organized according to topography. In this case, the smallest numbers of stations are associated with the smallest elevations and vice-versa.

The methods for generating the high-resolution temperature dataset are also described in Figure 2 with orange arrows.

## 2.4 | Climatologies

For providing a brief overview of the main climate features of the three countries, we computed seasonal means and annual cycle, with the 5-km gridded datasets. The annual cycle is calculated for each red box in Figure 1 and presented in heatmaps. These results will facilitate the comparison of our new dataset with others in the literature.

# 3 | RESULTS

## 3.1 | Quality control

As described in the methodology, consistency analysis involves objective procedures as well as subjective. In the objective procedure, the first step was to check if the meteorological stations were distributed across the studied domains. Despite the limited number of meteorological stations, Figures 3–5a demonstrates that these stations are spatially distributed across the three countries. However, it was noted that in Nepal, precipitation and temperature are not recorded from the same location in some stations. After removing all incorrect values from the time series,

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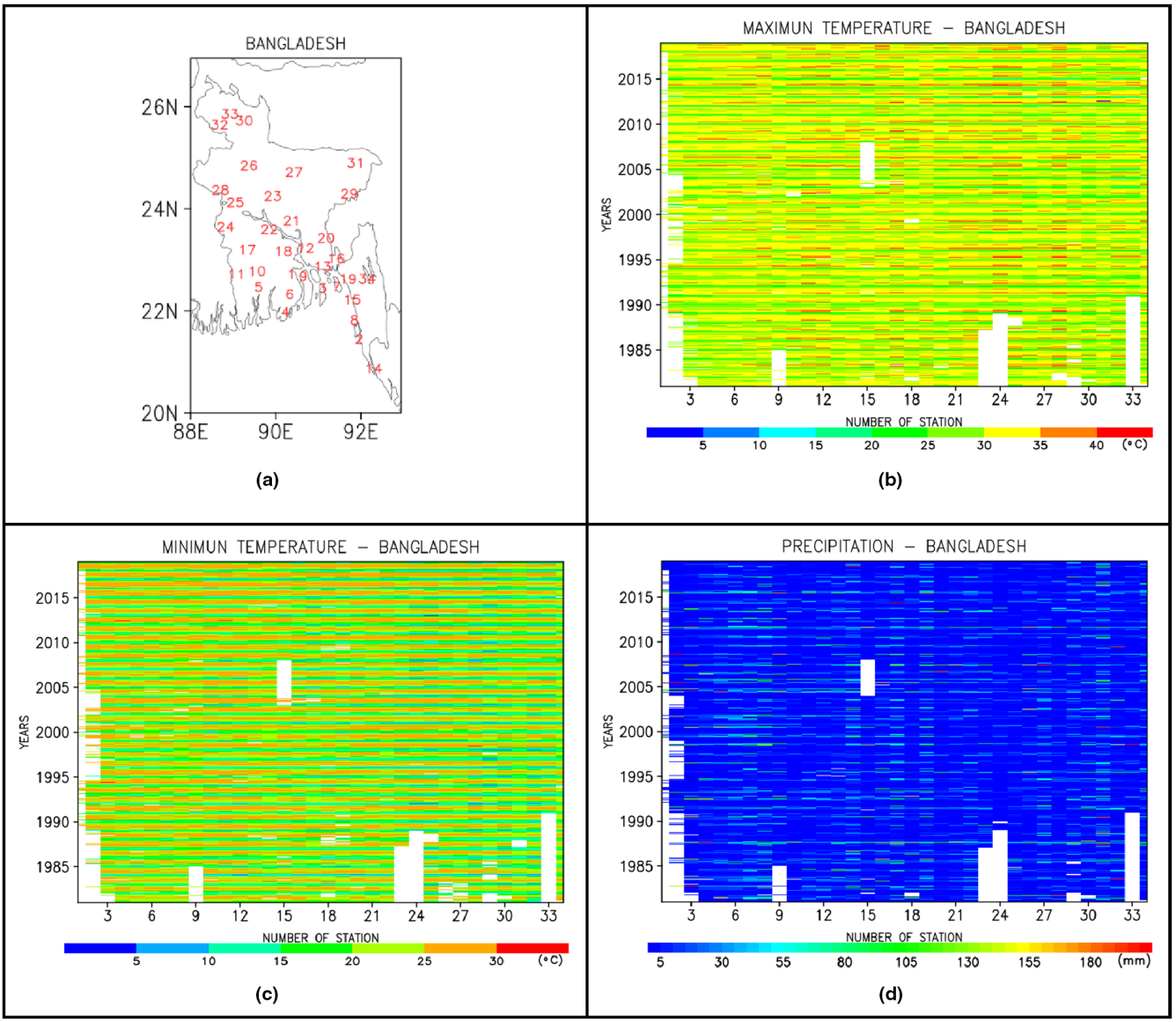



**FIGURE 4** Similar Figure 3 but to Bangladesh.

the number of valid data for each variable is presented in Table 1.

The daily time series for the entire period are also organized in a matrix for each country, with the columns representing the meteorological stations arranged according to altitude (topography). For example, in the case of Pakistan, there are 48 stations. The first station (number 1) is located near sea level, while the last station (number 48) is at the highest altitude (Figure 3b–d). This process allows us to identify the effect of topography on the variables. The matrices are presented in heat maps (Figure 3–5b-d) to facilitate the visualization of days with missing data (shown in white on the heat maps) and to observe the influence of the topography. Stations 5 and 43 for Pakistan show higher missing values for maximum temperature (Figure 3b). After quality control, Pakistan shows 97.4%, 97.2%, and 96.6% valid data for maximum temperature, precipitation, and minimum temperature, respectively (Table 1). Bangladesh has a higher percentage of valid data (92% for all three variables) compared to Nepal, which has 90%, 82.7%, and 81.1% for precipitation, maximum, and minimum temperature, respectively (Table 1).

For data uniformity in Pakistan, we utilized the station data from Pakistan Meteorological Department (PMD), which holds higher weight in the 5 km gridded dataset. However, there is a limitation regarding the availability of stations in northern Pakistan. This raises the question of the 5 km dataset performance. To address it, the air temperature measured in Khunjarab and Naltar meteorological stations from Water and Power Development Authority (WAPDA), not included in the 5 km data development, were used to validated the 5 km gridded dataset in that region. The performance of the 5 km gridded dataset is close to the stations

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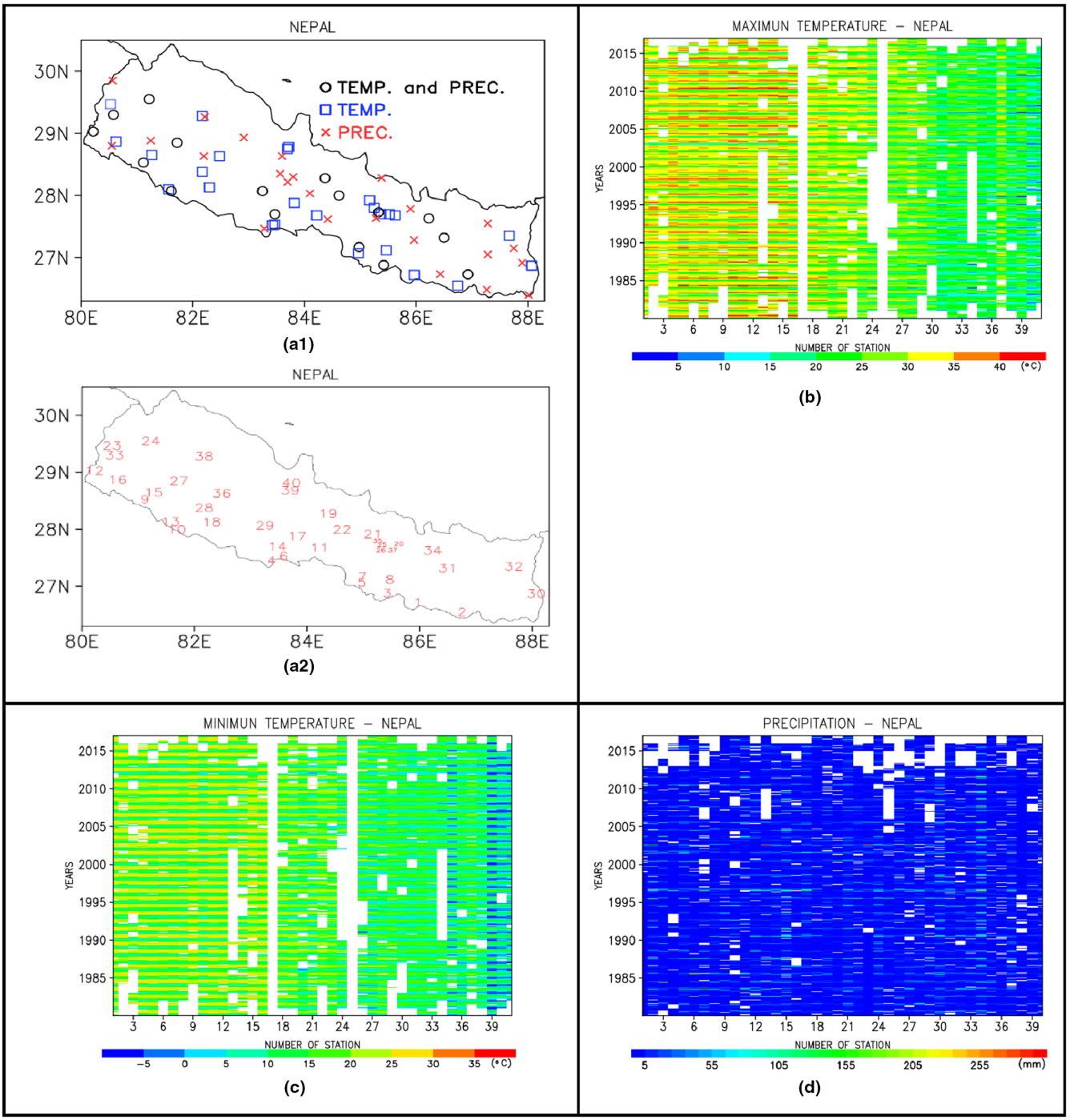



FIGURE 5 Similar Figure 3 but to Nepal. In (a1) black circles represent stations that measure both temperature and precipitation, blue squares represent stations that only measure temperature, and red crosses represent stations that only measure precipitation.

according different statistical analysis such as time correlation, which is higher than 0.94 (figures not shown).

## 3.2 | Developing seasonal climatology pattern estimates

For each country, Figure 6 shows the seasonal climatology of precipitation, minimum, mean, and maximum temperature, derived from the 5 km gridded dataset. In all three countries, precipitation is highest during the monsoon season (JJA) and lowest during DJF (Figure 6a–d). Nepal and Bangladesh are closer to the Bay of Bengal, where stronger winds transport higher moisture content towards the land (Nepal and Bangladesh) and eventually reach Pakistan (Webster et al., 1977). As a result, these countries receive more rainfall compared to Pakistan. Additionally, the topography plays

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**TABLE 1** The number of stations, percentage of valid data, and start and end of the data time series for Pakistan, Bangladesh, and Nepal.

| Countries | Number of stations | Valid data (%) | Started | End |
|---|---|---|---|---|
| Pakistan | 48 | Tmax 97.4<br>Tmin 96.6<br>Prec 97.2 | 1980/01/01 | 2017/12/31 |
| Bangladesh | 34 | Tmax 93.3<br>Tmin 92.1<br>Prec 93.2 | 1981/01/01 | 2018/12/31 |
| Nepal | 41 | Tmax 82.7<br>Tmin 81.1<br>Prec 90.0 | 1980/01/01 | 2016/12/31 |

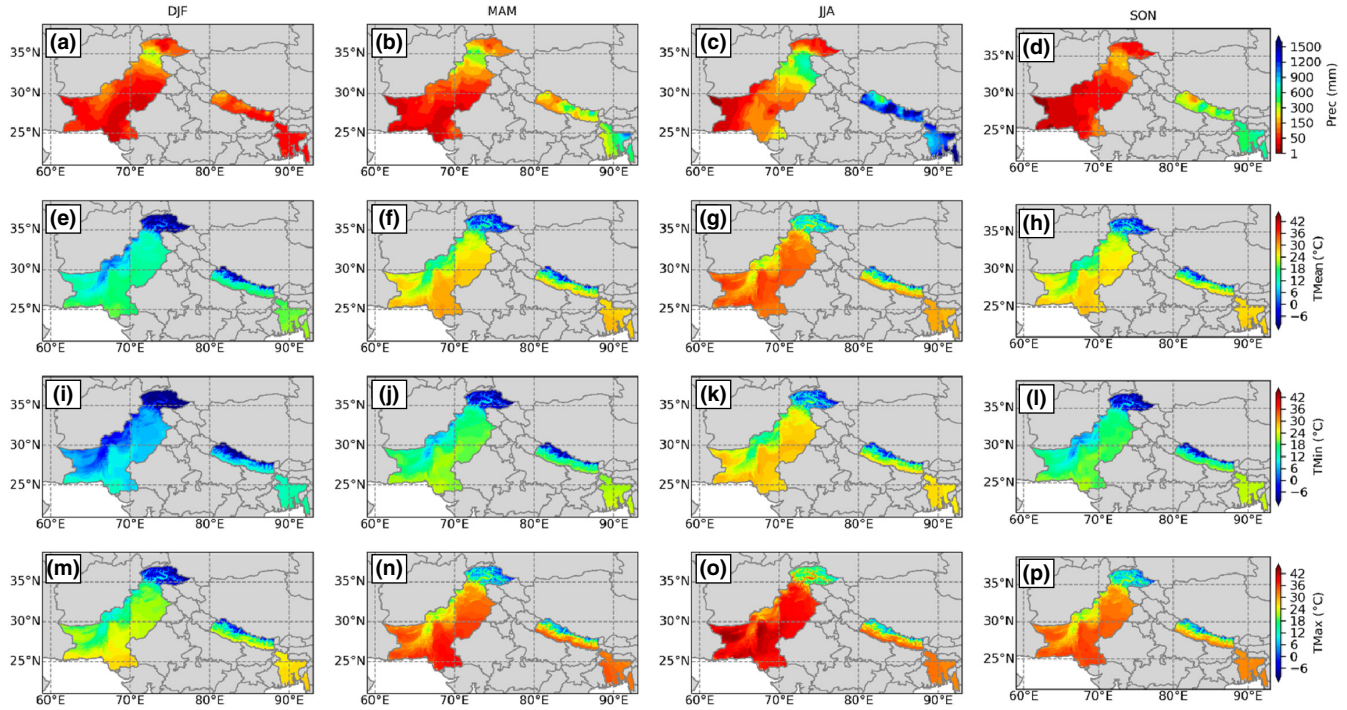



**FIGURE 6** Seasonal climatology of (a–d) precipitation (mm), (e–h) mean temperature (°C), (i–l) minimum temperature (°C), and maximum temperature (°C) for the period 1981–2016.

a significant role in Nepal, as the southern winds are forced to ascend when they reach the Himalayas, resulting in orographic precipitation.

The findings in Figure 7 depict the annual precipitation cycle estimated for the red box areas shown in Figure 1. Precipitation has higher seasonality in Nepal and Bangladesh compared to Pakistan. Furthermore, precipitation is higher in Bangladesh during June–July, while in Nepal it occurs 1 month later, specifically in July and August. This difference arises from the fact that the monsoon winds reach Bangladesh before they arrive in Nepal and subsequently in the Pakistan region P2 and the northern portion of P3, resulting in rainfall in July and August.

In terms of air temperature, Pakistan is the warmest among the three countries, and one of the reasons for this is the lower precipitation. In Pakistan, the solar energy is primarily used to heat the air instead of being utilized in evaporation processes. In contrast, Nepal and Bangladesh experience more rainfall, which helps maintain moderate temperatures (Figure 6).

The northern parts of Pakistan and Nepal show lower temperatures due to the latitude and altitude influences. On the other hand, Bangladesh has a slight spatial variability in air temperature since it is a plain region (topography does not have too many variations, Figure 1). In addition, the difference between the minimum and maximum temperature is high for Pakistan, while it has a smaller range in Bangladesh due to the maritime influence (Figure 6). It is also shown in Figure 8, which presents the annual cycle of mean temperature.

Results in Figure 8 indicate that the coldest area is P1, located in northern Pakistan, while P4 also exhibits low

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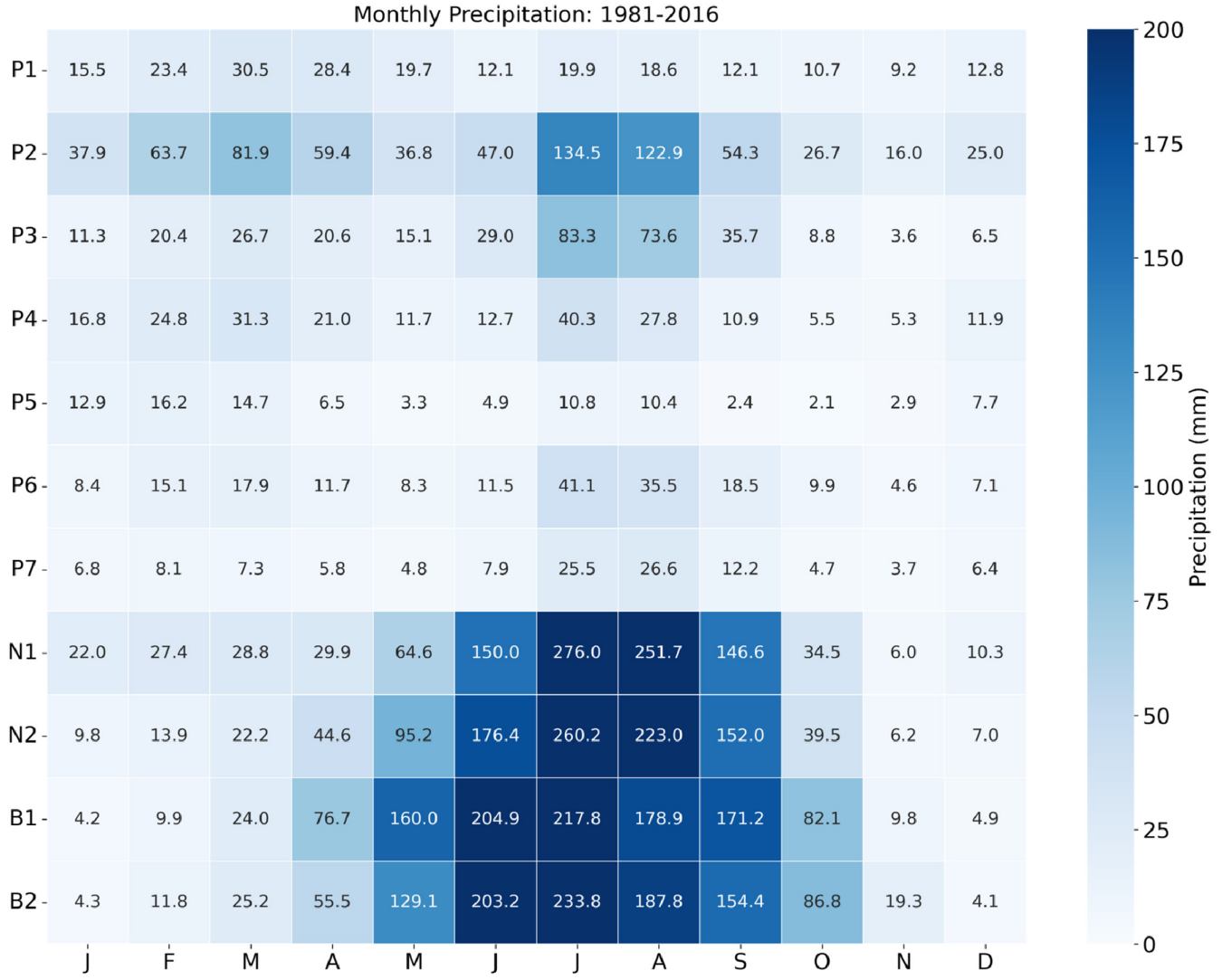



**FIGURE 7** The monthly mean precipitation (mm) during 1981–2016 for each region shown in Figure 1.

temperatures during DJF. The northern side of Nepal, influenced by the Himalayan mountains, also experiences low temperatures. Bangladesh does not have great variability in the temperature across its regions throughout the year. Among the three countries, Pakistan has both the hottest region (P7) and the coldest region (P1), in comparison to Bangladesh and Nepal.

## 4 | CONCLUSION

Knowledge of local-scale climate features is crucial for economic activities, development, adaptation, and mitigation of climate change. However, observational data are scarce in many regions of South Asia, particularly in mountainous areas. To overcome this data limitation, we prepared high-resolution daily gridded data with a 5-km resolution for Bangladesh, Nepal and Pakistan, from 1981 to 2016.

Although there are few meteorological stations in the three countries, their distribution in these regions enables for the observation of various atmospheric conditions. However, extreme north Pakistan and Nepal have a limited number of stations due to their high elevations, reaching approximately 8,000 m.

A data quality test for meteorological stations used in this study reveals that Pakistan has the highest percentage of valid data (~96%), followed by Bangladesh (93%) and Nepal (less than 90%). To develop a high-resolution 5-km dataset, we merged the station datasets with ERA5 reanalysis. Additionally, a process was implemented to account for topography in temperature estimation by considering a vertical LRT using a high-resolution digital elevation model. This process was applied only in Pakistan and Nepal because Bangladesh has a flat topography.

Comparing the climatology of the three countries, Pakistan is generally the warmest, and Bangladesh is the

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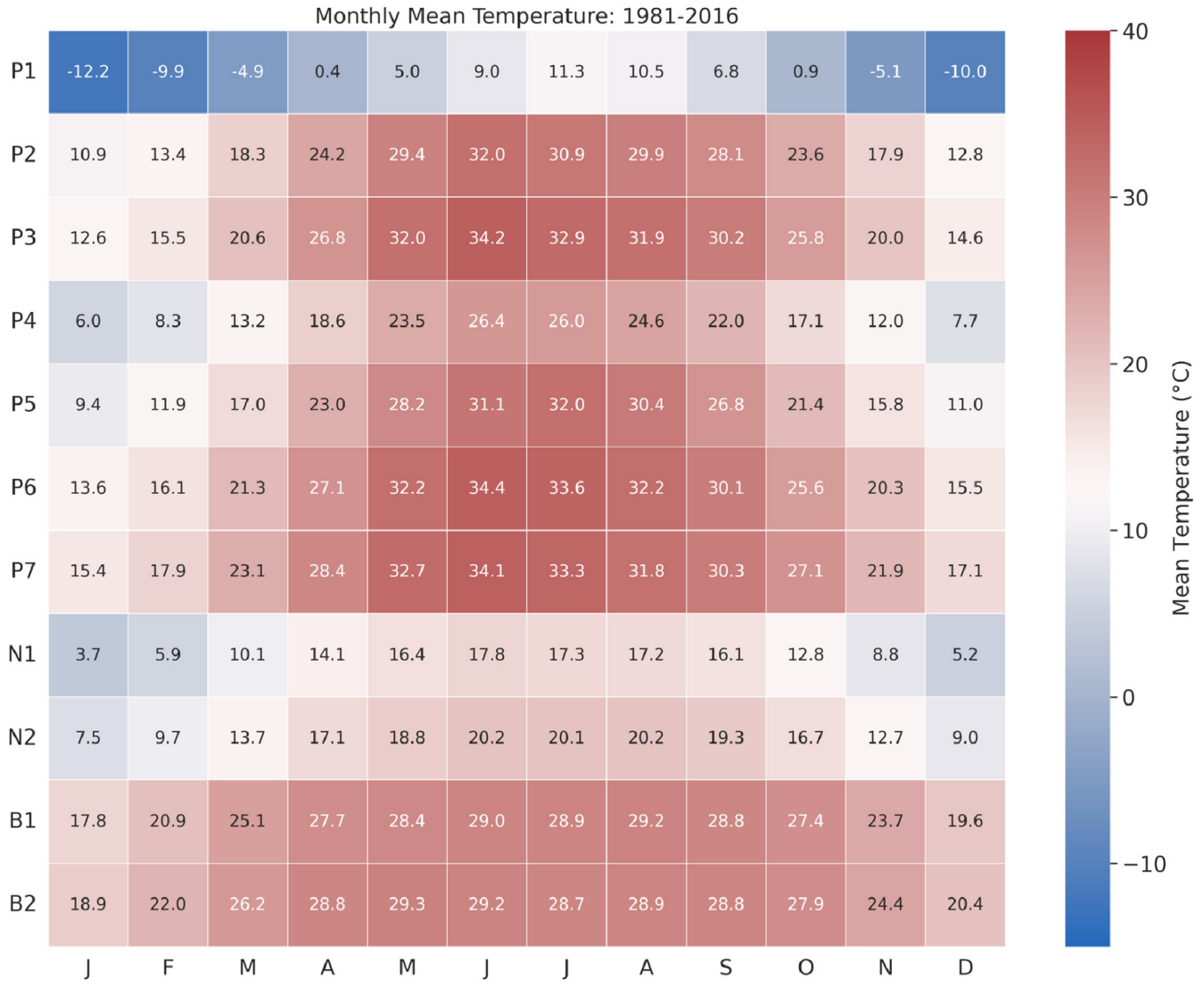



**FIGURE 8** The monthly mean temperature (°C) during 1981–2016 for each region shown in Figure 1.

wettest. The monsoon circulation influences the seasonal cycle of precipitation in these countries.

The high-resolution daily data product can be useful for impact studies such as extreme events, water resources, agriculture, health, energy, etc. Moreover, this fine-resolution data will be utilized to downscale CMIP6 model data for analysing future climate and its impacts in Pakistan at a higher resolution, and also to assist in local policy decisions.

## AUTHOR CONTRIBUTIONS

**Shaukat Ali:** Conceptualization; project administration; supervision. **Zulfiqar A. Bhutta:** Resources. **Muhammad Arif Goheer:** Resources. **Shiva Ebrahimi:** Methodology. **Jose Roberto Rozante:** Methodology; investigation; writing – original draft; validation; visualization; funding acquisition; software. **Sher Muhammad:** Writing – original draft. **Firdos Khan:** Writing – original draft. **Rida Kiani:** Data curation; writing – original draft. **Li Dan:** Writing – original draft.

## ACKNOWLEDGEMENTS

This work is supported by the project of Asia-Pacific Network for Global Change Research (APN) titled "Towards robust projections of climate extremes and adaptation plans over South Asia: CRRP2018-04MYAli". The authors also thank all centres that provided the data used in the study. We also get technical support from the Cryosphere Initiative of the International Centre for Integrated Mountain Development (ICIMOD).

## FUNDING INFORMATION

This work is supported by the project of Asia-Pacific Network for Global Change Research (APN) titled "Towards robust projections of climate extremes and adaptation plans over South Asia: CRRP2018-04MYAli".

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## CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

## DATA AVAILABILITY STATEMENT

The 5-km gridded dataset of precipitation and minimum and maximum temperatures for Bangladesh, Nepal, and Pakistan, from 1981 to 2016, are publicly available and can be download from https://doi.org/10.6084/m9.figshare.21565149.v1 (Ali & Reboita, 2022).

## OPEN RESEARCH BADGES

This article has been awarded Open Data Badge for making publicly available the digitally-shareable data necessary to reproduce the reported results. Data is available at https://doi.org/10.6084/m9.figshare.21565149.v1

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**How to cite this article:** Ali, S., Bhutta, Z.A., Reboita, M.S., Goheer, M.A., Ebrahimi, S., Rozante, J.R. et al. (2024) A 5-km gridded product development of daily temperature and precipitation for Bangladesh, Nepal, and Pakistan from 1981 to 2016. *Geoscience Data Journal*, 11, 292–302. Available from: https://doi.org/10.1002/gdj3.217

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